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Review article

Modeling and optimization of carbon capture, utilization, and storage networks: A review of mathematical approaches

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ABSTRACT

This paper presents a systematic literature review (SLR) on mathematical modeling for carbon capture and storage (CCS) and carbon capture, utilization, and storage (CCUS) supply chain network design. The main objective is to examine how mathematical programming methods have been employed to model CCS/CCUS systems by exploring the modeling scope, identifying key network structures, analyzing the primary objectives addressed, and reviewing solution methods used in case studies. Relevant studies were retrieved from Scopus and Web of Science using defined search strings. The review provides a structured classification of models, distinguishing between single-objective and multi-objective formulations. It also categorizes models based on structural design, including two-echelon and multi-echelon frameworks. Additionally, models are classified as deterministic or non-deterministic, with the latter incorporating various forms of uncertainty. Finally, based on this critical classification of mathematical models, along with the modeling scope and current insights, the review highlights key areas for future research to enhance the comprehensiveness and practical relevance of CCS/CCUS modeling.

1. Introduction

Climate change is primarily driven by the increasing concentration of greenhouse gases (GHGs), including CO_2, CH_4, N_2O, and fluorinated gases, leading to severe environmental consequences such as rising global temperatures, rising sea levels, and more frequent natural disasters (United Nations, 2022; EPA, 2023; National Academy of Sciences, 2020; Ebi et al., 2020; Abbass et al., 2022). The greenhouse effect intensifies global warming by trapping solar radiation, with CO_2 emissions from fossil fuel combustion accounting for approximately 60% of total GHG emissions, while CH_4 contributes around 18% (National Academy of Sciences, 2020; Abbass et al., 2022; Malhi et al., 2020). Global CO_2 emissions increased from 34.65 billion metric tons in 2013–37.01 billion metric tons in 2023, representing a 6.8% rise over the decade (Statista, Global Carbon Budget, 2024). Power generation remains the largest source of CO_2 emissions, producing approximately 14.65 billion tons of CO_2 from 28,000 TWh of electricity in 2022 (International Energy Agency, 2023; Ritchie and Rosado, 2020; IEA, 2023; Enerdata, 2023). Coal-fired power plants, gas-fired power plants, and combined-cycle power plants play a dominant role due to their high direct CO_2 emission factors, with coal-fired power plants emitting 760 kg of CO_2 per MWh and combined-cycle power plants emitting 370 kg of CO_2 per MWh (IPCC, 2014). Additionally, the transportation, manufacturing, and construction sectors, particularly cement production and similar industries, are also major contributors to global CO_2 emissions (Ritchie and Rosado, 2020).

One approach to mitigating CO_2 emissions is through Carbon Capture and Storage (CCS) or Carbon Capture, Utilization, and Storage (CCUS). These technologies allow fossil fuel operations to continue while still meeting CO_2 reduction targets for climate change mitigation (Bui et al., 2018). CCS focuses on capturing and storing CO_2 to prevent its release into the atmosphere. In contrast, CCUS includes CO_2 utilization, converting it into value-added products as an alternative to storage (Valluri et al., 2022; Hong, 2022). Global CCS capacity has expanded significantly in recent years. It increased from 85 million tons per year in 2019–110 million tons per year in 2020 and reached 149 million tons per year in 2021 (Page et al., 2019, 2020; Turan et al., 2021). Power generation projects, particularly in the Americas and Europe, contribute the largest share of global CO_2 capture capacity. As of September 2021, these projects accounted for 37% of the annual global total, with a

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capacity of 62.5 million tons of CO_2 per year (Hong, 2022).

CCUS technology involves four main processes: CO_2 capture, compression, transportation, and either storage or utilization. Capture methods include direct air capture, post-combustion, pre-combustion, oxy-combustion, and chemical looping. Post-combustion is widely used due to its high capture efficiency (80–90%) and adaptability for retrofitting existing coal power plants, with proven small-scale commercial use (Dubey and Arora, 2022; Rubin et al., 2012; Wilberforce et al., 2021). Captured CO_2 is compressed into a supercritical state ($\approx$15 MPa) for efficient pipeline transport, which is cost-effective over long distances and for large volumes (Leung et al., 2014; Zhang et al., 2018a). Alternative transport (ship, truck, train) requires cryogenic cooling. Water is removed pre-transport to prevent corrosion (Hong, 2022; Kearns et al., 2021). CO_2 can be utilized in chemical production (e.g., methanol, urea) or stored in saline formations and depleted oil/gas fields. It also supports enhanced oil/gas recovery (EOR/EGR) and enhanced coal bed methane (ECBM), where injected CO_2 aids extraction and remains stored underground (Hong, 2022; Kearns et al., 2021).

Supply Chain Network Design (SCND) is a strategic process that involves planning and optimizing the structure of supply chains. It encompasses critical decisions regarding the location and capacity of facilities, material flow, and resource allocation to meet demand efficiently while minimizing costs (Zanjirani et al., 2014; Govindan et al., 2017). In operations research, SCND is employed to determine optimal facility placement, capacity planning, and material distribution within a network (Eskandarpour et al., 2015).

To represent these complex systems, mathematical models are commonly used. These models abstract real-world systems using mathematical language—symbols, equations, and variables—to describe the relationships between system components. They help visualize interactions and enable the computation of unknown variables (Mirhassani and Hooshmand, 2019; Thie and Keough, 2008; AIMMS, 2023).

One of the core tools for SCND is mathematical programming, particularly Linear Programming (LP). LP solves optimization problems by formulating objective functions and constraints using linear equations (Griva et al., 2009; Rashid et al., 2017). Its versatility has led to broad applications across economics, engineering, logistics, and transportation. Mixed-Integer Linear Programming (MILP) extends LP by including integer variables, allowing it to model complex decision-making problems such as scheduling and resource allocation (Enrique et al., 2001).

In the context of CCS/CCUS, mathematical programming is one of the most widely applied methods for solving SCND problems. Other commonly used approaches include pinch analysis, numerical simulation, and P-graph modeling. LP is particularly favored due to its ability to manage multiple constraints simultaneously. Nevertheless, it also faces certain limitations, such as difficulty in achieving global optima in non-linear models, long computation times for large-scale systems, and interpretability challenges in multi-objective problems where multiple solutions may be equally valid (Frederick et al., 2017, 2016a, 2016b).

Many review articles have been conducted discussing the development of CCUS technologies, their energy requirements, costs, and implementation challenges. Hong provides a review of CCUS systems focusing on technology readiness, technical performance, energy requirements, and associated costs (Hong, 2022). Orujov et al. review technical progress in CO_2 capture, utilization, storage, and monitoring, along with regulatory and policy developments globally. They highlight methods to enhance storage capacity—such as foam injection and nanoparticle stabilization—and emphasize safety and monitoring strategies for long-term geological storage (Orujov et al., 2023). Liu et al. review the current state and challenges of CCUS, highlighting advancements in CO_2 capture (e.g., ionic liquids, metal–organic frameworks), transport safety, utilization technologies, and geological storage. They identify key barriers such as low capture efficiency, high costs, and weak policy and international collaboration that hinder broader CCUS deployment (Liu et al., 2023). Chen et al. identifies CCUS as crucial for carbon neutrality but highlights significant challenges. These include financial, technical, and regulatory risks, as well as uncertainties in geological storage and project success. Their study defines the optimal expansion period as 2040–2060 globally and 2030–2050 in China (Chen et al., 2022). Jiang et al. highlight CCUS as essential for climate change mitigation and a low-carbon future. Despite its inclusion in China's national strategy since the 12th Five-Year Plan, the policy framework remains weak. Key issues include insufficient legal structures, limited operational guidance, lack of market incentives, and inadequate financial subsidies (Jiang et al., 2020). Mahjour and Faroughi (2023) highlight CCS as capable of reducing industrial emissions by 25–67%. Their study provides recommendations to improve feasibility, security, and cost-efficiency (Mahjour and Faroughi, 2023). Tapia et al. review CCUS as a key climate mitigation strategy. They focus on CO_2 capture for utilization and long-term storage. Their study explores Process Systems Engineering (PSE) advancements for large-scale CCUS planning. They identify research opportunities to enhance CCUS integration and improve decision-making tools (Frederick et al., 2017). Meanwhile, Tian et al. reviewed various types of CCUS models, including engineering models, economic models, and techno-economic models. Their study also examined modeling approaches in CCS/CCUS that adopt both single-objective and multi-objective optimization frameworks (Tian et al., 2016). However, the review by Tian et al. lacks comprehensiveness due to the limited number of articles analyzed.

This article distinguishes itself from previous reviews by specifically focusing on the development and application of mathematical modeling in the design and optimization of CCUS supply chain networks. While earlier studies have primarily explored technological advancements, techno-economic assessments, policy frameworks, and decision-support tools, this review addresses a critical gap by examining how mathematical programming methods have been employed to model, analyze, and optimize CCUS systems. The main objective is to provide a structured overview of recent progress in mathematical modeling within CCUS supply chains, emphasizing its role in enhancing system efficiency, cost-effectiveness, and strategic planning. The scientific contribution of this review lies in its targeted synthesis of optimization approaches, its systematic classification of modeling techniques, and its identification of key research gaps—particularly in the modeling of uncertainty.

This article is structured to provide a comprehensive overview of CCUS supply chain network design using mathematical modeling. Section 1 introduces the study's background and objectives. Section 2 covers the development of CCUS technologies, focusing on their *Technology Readiness Level* (TRL). Section 3 examines the application of mathematical models to supply chain design challenges. Section 4 describes the *Systematic Literature Review* (SLR) methodology used to collect and analyze studies. Section 5 reviews recent advancements in mathematical modeling for CCUS, identifies research gaps, and suggests future research directions based on the SLR findings. Section 6 concludes the paper.

2. CCUS technology

Fig. 1 illustrates the main stages of the CCU/CCUS process, which include carbon capture, CO_2 transportation, utilization, and storage. The selection of appropriate carbon capture technology depends on the characteristics of the emission source, which may include natural gas processing, steel manufacturing, cement production, power generation, ammonia synthesis, hydrogen production, and fermentation. In particular, hydrogen production and fermentation generate nearly pure CO_2 streams, making them highly suitable for efficient capture and utilization in CCUS systems (Metz et al., 2015). Once captured, CO_2 is transported to either storage sites or utilization facilities using various transportation modes (Page et al., 2020). The most common and efficient method is pipeline transport, where CO_2 is compressed into a

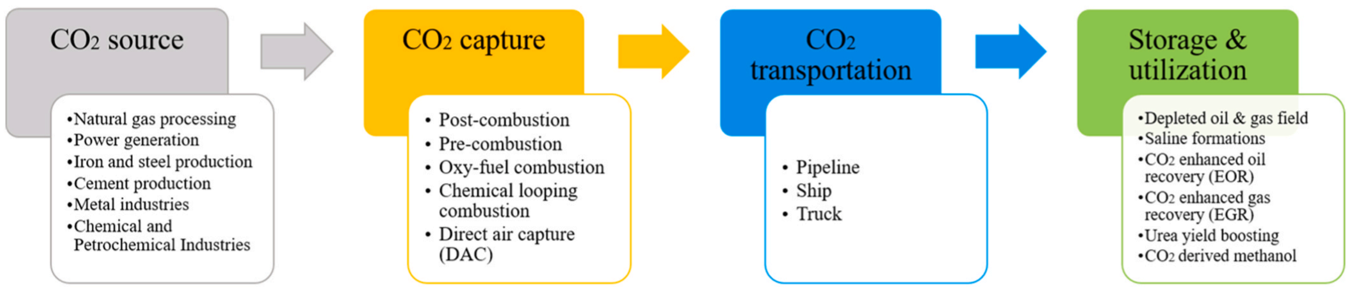

Fig. 1. Process chain in CCUS technology.

supercritical state—enhancing transport efficiency and reducing costs, especially for large volumes and long-distance transfer from sources expected to operate for over 20 years (Leung et al., 2014; Zhang et al., 2018a). Alternative transportation options such as ships, trucks, and trains require CO_2 to be cooled into its liquid phase to ensure stability during transport. Prior to transportation, water is removed to prevent pipeline corrosion, and CO_2 is typically compressed to around 15 MPa to maintain its supercritical condition (Hong, 2022; Kearns et al., 2021).

There are several CO_2 separation technologies, including absorption, adsorption, membrane, cryogenic, and biological methods, as shown in Fig. 2. Absorption is the most mature and commercially adopted method for separating CO_2 from flue gases, especially in the oil and chemical industries. It involves liquid solvents that can be regenerated through heating or decompression (Leung et al., 2014; Wang and Song, 2020). Absorption is divided into chemical and physical types: chemical absorption relies on acid-base reactions between CO_2 and alkaline solvents, while physical absorption depends on CO_2 solubility under varying temperature and pressure conditions (Songolzadeh et al., 2014; Yuan and Eden, 2016). Common chemical solvents include MEA, AMP, DEA, and MDEA, whereas physical solvents include DEPG (Selexol), chilled methanol (Rectisol), NMP (Purisol), and propylene carbonate (Fluor). Adsorption involves capturing CO_2 on the surface of solid materials and is widely used in industrial applications and power plants for pre-combustion, post-combustion, and direct air capture (DAC) (Bui et al., 2018; Leung et al., 2014).

Membrane separation uses permeable or semi-permeable materials to isolate CO_2 based on mechanisms like solution-diffusion, molecular sieving, and Knudsen diffusion (Yuan and Eden, 2016). Cryogenic separation compresses and cools gas mixtures to very low temperatures (–100 °C to –135 °C) and high pressures (101–203 bar) to liquefy and separate high-purity CO_2 (99.99%). This method can be applied in both pre- and post-combustion capture systems (Leung et al., 2014; Song et al., 2019; Mondal et al., 2012). Biological CO_2 capture uses plants and microorganisms—such as bacteria, algae, fungi, and yeast—to remove CO_2 from the atmosphere or flue gas. Common biological approaches include afforestation, ocean fertilization, and biophotolysis using cyanobacteria or microalgae (Yang et al., 2008).

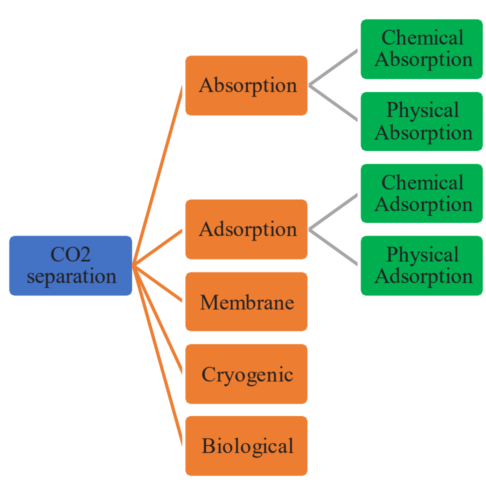

Fig. 2. CO_2 separation technology.

Each CO_2 emission source has a distinct flue gas composition that influences CO_2 separation strategies. High CO_2 concentrations are found in agricultural processing, cement, and sugar industries (47%), and in iron and steel processes (44%). Moderate concentrations occur in gas processing (25%), iron and steel (24%), and industrial sectors such as aluminum and glass (around 15%). Coal-based power plants emit flue gas with approximately 14% CO_2, while natural gas-based power plants and compressor stations emit much lower levels, around 4%. In general, the higher the CO_2 concentration in the flue gas, the lower the cost of capture and compression (Hasan et al., 2014).

Hasan et al. found that MEA-based chemical absorption is the most cost-effective option at low CO_2 concentrations (<15–20%), while vacuum swing adsorption (VSA) becomes more economical at higher concentrations (Hasan et al., 2014). Adsorption methods using PSA and VSA with 13X zeolite consistently maintain low costs across a wide CO_2 range, achieving over 90% capture efficiency and compression up to 150 bar (Hasan et al., 2012a). In contrast, membrane technologies are the most expensive and least sensitive to CO_2 concentration variation, based on comprehensive modeling and optimization studies (Hasan et al., 2012b). Building on Hasan's work, Kalyanarengan et al. developed a linear model characterizing the cost behavior of various carbon capture technologies based on CO_2 concentration (Kalyanarengan et al., 2017).

CO_2 separation technologies each have distinct strengths and weaknesses. Absorption is the most mature and widely used, offering high efficiency (>90%) and low-cost solvent use, but it is energy-intensive, sensitive to contaminants (e.g., SO_x, NO_x), and less effective at high temperatures and low pressures. Adsorption provides high efficiency (>85%), low corrosion, and reusability of adsorbents, but performance declines in the presence of moisture and at higher temperatures, and chemical adsorbents require high regeneration energy. Membrane systems offer ease of operation and relatively low operational costs, though they suffer from high manufacturing costs, low selectivity, and fouling—especially in moist conditions. Cryogenic separation is efficient (>85%) and established, but it is highly energy-intensive and requires pre-removal of moisture. Biological methods are eco-friendly and can produce valuable byproducts, yet they are slow, land-intensive, and sensitive to environmental variables, with lower CO_2 removal efficiency (Hong, 2022; Leung et al., 2014; Metz et al., 2015; Wang and Song, 2020).

Technology Readiness Level (TRL) is a framework for assessing technology maturity, guiding its progression from concept development to real-world application (Hirshorn and Jefferies, 2016; European Commission, 2014). In the CCUS TRL model, technology evolves through three phases. The research phase (TRL 1–3) involves fundamental studies and initial proof-of-concept work. The development phase (TRL 4–6) includes prototype validation and testing in controlled environments. The demonstration phase (TRL 7–9) focuses on large-scale implementation and operational deployment (Hong, 2022). Some CCUS technologies with TRL 7–9 include: (Bui et al., 2018; Kearns, Liu, and Consoli, 2021):

- At TRL 7 (Demonstration), several CO_2 capture technologies are being tested at an advanced stage, including polymeric membranes for the natural gas industry, pre-combustion IGCC with CCS, oxy-combustion coal power plants, and post-combustion temperature swing adsorption (TSA). Key CO_2 storage technologies at this stage include BECCS (Bioenergy with Carbon Capture and Storage), direct air capture (DAC), depleted oil and gas fields, and CO_2-enhanced gas recovery (EGR).
- At TRL 8 (Commercial Refinement Required), CO_2-derived methanol is undergoing further refinement for commercial applications.
- At TRL 9 (Commercial Deployment), several CO_2 capture technologies such as post-combustion amines for power plants, pressure swing adsorption (PSA)/vacuum swing adsorption (VSA), and pre-combustion natural gas processing are fully implemented. CO_2 transport is well-established through onshore and offshore pipelines and ships. In CO_2 utilization, urea yield boosting has been successfully deployed. CO_2 storage technologies, including saline formations and CO_2-enhanced oil recovery (CO_2-EOR), are widely used for long-term carbon sequestration

Several CCUS technologies with TRL 7–9 are advancing. Post-combustion capture is widely used and retrofits easily but has low efficiency and high energy costs. Pre-combustion capture improves CO_2 concentration but faces efficiency losses and high operational costs. Oxy-fuel combustion enhances CO_2 absorption but suffers from efficiency drops and expensive O_2 production. Chemical looping combustion (CLC) eliminates costly air separation but remains under development and requires fuel desulfurization. Direct Air Capture (DAC) removes atmospheric CO_2 but demands high energy due to low CO_2 concentrations (Hong, 2022; Leung et al., 2014).

At 1 bar and 20°C, CO_2 is a gas, making flow measurement easy. At 57 bar and 20°C, it changes to a liquid state. Above 31.1°C and 73.9 bar, CO_2 becomes supercritical, with liquid-like density and gas-like compressibility and low viscosity (Simonsen et al., 2024). Since 31.1°C is close to ambient temperature in many regions, CCUS operations often approach this critical point, where small variations in temperature and pressure can significantly affect fluid properties and complicate process control and measurement accuracy. The CO_2 phase diagram, illustrating these transitions, highlights the expected CCUS operating range and its impact on measurement, transport, and storage processes (Mills et al., 2022). CO_2 transportation is a crucial part of the CCS chain, linking CO_2 sources to storage sites via pipelines, ships, trucks, or trains. Captured CO_2 is transported in large volumes by compressing it into a supercritical state. For pipeline transport, CO_2 remains supercritical, while for ship, truck, or rail transport, it is cooled into a liquid state (as shown in Fig. 3). Before transport, H_2O in captured CO_2 is removed to prevent acid formation and corrosion in pipelines and equipment. Dehydration is typically performed alongside compression or cooling.

CO_2 can be utilized or stored, with several utilization technologies already demonstrated at advanced scales (TRL 7–9), including CO_2-derived methanol and urea yield boosting (Bui et al., 2018; Kearns et al., 2021). Methanol production from CO_2 can be achieved through methods such as catalytic hydrogenation or electrochemical conversion, with catalytic hydrogenation already implemented at industrial scale (Goeppert et al., 2014). Concentrated CO_2 is hydrogenated to synthesize methanol (CH_3OH) over metal or metal oxide catalysts, typically copper (Cu), zinc oxide (ZnO), and alumina (Al_2O_3), at moderate temperatures and pressures. Hydrogen is produced via water electrolysis, making this process energy-intensive, but it becomes environmentally friendly and economically feasible if powered by renewable energy sources (Patricio et al., 2017). For urea production, CO_2 is captured from industrial flue

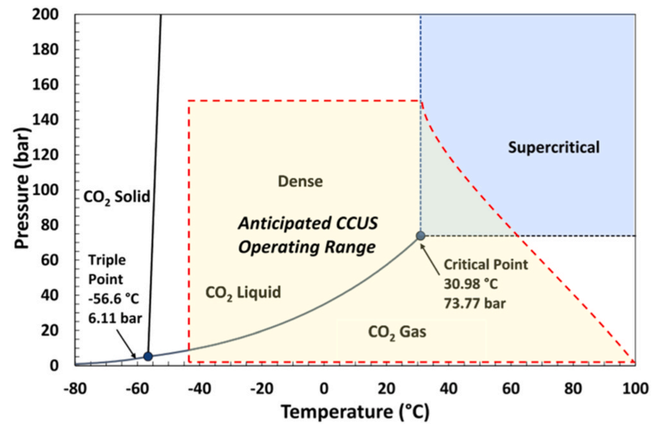

Fig. 3. The pure CO_2 phase diagram highlights the CCUS operating range in yellow (Mills et al., 2022).

gases using technologies such as monoethanolamine (MEA) solutions or chilled ammonia processes (CAP). The captured CO_2 is reacted with ammonia, forming ammonium carbamate, which subsequently converts into urea through dehydration (Shirmohammadi et al., 2020; Chehrazi and Moghadas, 2022).

CO_2 utilization has been employed to enhance oil and gas production through CO_2-enhanced oil recovery (TRL 9) and CO_2-enhanced gas recovery (TRL 8)) (Kearns et al., 2021). In CO_2-EOR, supercritical CO_2 is injected into nearly depleted oil reservoirs, mixing with oil to lower viscosity. This reduces the interfacial tension between oil and gas, improving oil flow through reservoir rock pores toward production wells without increasing well pressure (Simonsen et al., 2024; Asian Development Bank, 2019). Carbon dioxide is injected into oil reservoirs due to its ability to become miscible with oil under reservoir conditions (see Fig. 4). This miscibility causes the oil to swell and reduces its viscosity, enabling easier flow within the reservoir (Abdullah and Hasan, 2021). CO_2 storage is also feasible through enhanced gas recovery (EGR), where compressed CO_2 is injected into porous rock formations containing natural gas, increasing reservoir pressure. Over time, CO_2 partially dissolves into formation fluids and becomes mineralized, gradually reducing pressure and allowing additional CO_2 storage. Furthermore, the injected CO_2 helps push natural gas toward production wells, with some CO_2 being recycled and reinjected (Liu et al., 2022; Davoodi et al., 2023).

Captured CO_2 can be permanently stored in depleted oil and gas fields (TRL 7) or saline formations (TRL 9) (Kearns et al., 2021). Depleted reservoirs offer secure, cost-effective storage, proven by their historical retention of hydrocarbons. They have extensively studied geology, established infrastructure, and existing predictive modeling capabilities, with storage capacities between 675 and 900 $GtCO_2$, potentially rising by 25% if undiscovered reservoirs are (Turan et al., 2021; Celia et al., 2018; Agartan et al., 2018). Saline formations provide the largest potential, estimated at 1000–10,000 $GtCO_{22}$ (Celia et al.,

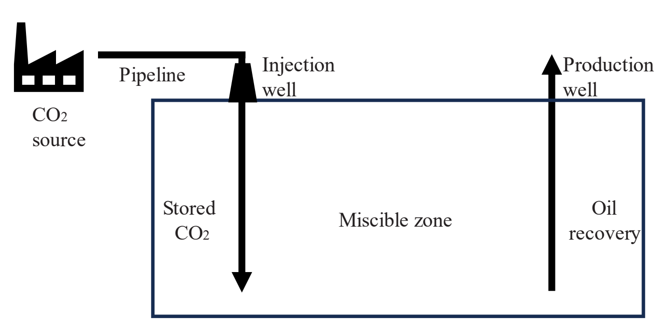

Fig. 4. Enhanced oil recovery through miscible CO_2 flooding.

2018). CO_2 trapping mechanisms here include hydrodynamic trapping beneath low-permeability caprocks, residual trapping in pore spaces, solubility in briny water, and mineralization. However, storage in saline formations poses challenges such as uncertainties in sealing integrity, interactions with dissolved minerals, leakage through geological faults, and induced seismic activity from high injection rates (Hong, 2022; Davoodi et al., 2023; Riaz and Cinar, 2014).

3. Supply chain network design of CCUS

3.1. Overview of SCND

Supply Chain Network Design (SCND) is a strategic process that defines the physical and operational structure of supply chains, including the location, number, and capacity of facilities, as well as material flow between them (Govindan et al., 2017; Chalmardi and Camacho-Vallejo, 2019). Operations research applies SCND to optimize these elements for cost-efficiency and responsiveness (Eskandarpour et al., 2015).

SCND operates at three decision-making levels—strategic, tactical, and operational, as follows (Zanjirani et al., 2014; Osmani and Zhang, 2017; Madani and Saihi, 2024):

- Strategic decisions involve long-term factors such as supplier selection, facility siting, and logistics planning.
- Tactical decisions focus on resource allocation, capacity planning, and pricing.
- Operational decisions address day-to-day activities such as inventory control and scheduling.

Given the complex and uncertain environment in which supply chains operate, SCND is essential for enhancing resilience and ensuring business continuity (Maharjan and Kato, 2022). Moreover, sustainability-oriented SCND models simultaneously address economic, environmental, and social objectives (Zanjirani et al., 2014; Eskandarpour et al., 2015).

In the context of CCUS, SCND supports the matching of CO_2 sources with utilization or storage sites, the design of cost-effective transportation routes, and the evaluation of trade-offs among economic, environmental, and social impacts (Frederick et al., 2017; Fan et al., 2021; Huang et al., 2013).

3.2. Mathematical modeling approaches in SCND

A model represents a system that captures specific aspects of reality to enhance understanding and support decision-making. Based on how it represents reality, models can be classified into four main categories: iconic, analog, verbal, and mathematical. A mathematical model is an abstract representation expressed through mathematical language—symbols, numbers, and equations—that describes relationships among system components, either descriptively or for estimating unknown variables (Mirhassani and Hooshmand, 2019; Thie and Keough, 2008; AIMMS, 2023).

Mathematical modeling plays a central role in SCND. These models commonly employ binary variables for facility location, technology selection, and transport mode, while continuous variables represent material or product flows (Eskandarpour et al., 2015). They are typically formulated as Mixed-Integer Linear Programming (MILP) or Mixed-Integer Nonlinear Programming (MINLP) problems.

Deterministic models rely on fixed parameters, whereas stochastic models incorporate uncertainty through probability distributions (Al-haidous and Al-ansari, 2020). Objective functions may be single-objective (SO)—for example, minimizing total cost—or multi-objective (MO), integrating additional dimensions such as environmental performance, typically measured in terms of Global Warming Potential (Madani and Saihi, 2024; Zhang et al., 2020a). This general formulation is illustrated in Fig. 5.

3.3. Modeling Frameworks for CCUS Networks

In CCUS applications, two main mathematical modeling frameworks are commonly adopted:

1. Source–Sink Matching (Single-Echelon Models):

This simplified framework involves two types of nodes—CO_2 emission sources and storage sites—and aims to minimize total transportation costs based on distance (Fig. 6).

2. Multi-Echelon CCUS Supply Chain Networks: This more comprehensive framework integrates CO_2 sources, capture technologies, transportation infrastructure, and sinks (Fig. 7). It accounts for variations in flue gas composition that influence technology selection, capture efficiency, and material requirements. The multi-echelon configuration allows for more realistic network representation and system-wide optimization.

3.4. Solution methods and optimization techniques

To solve complex SCND problems, Integer Programming (IP) and Mixed-Integer Linear Programming (MILP) approaches are widely used (Fávero and Belfiore, 2019).

- IP restricts all decision variables to integers.
- MILP combines discrete and continuous variables, making it particularly suitable for problems involving facility location, scheduling, and network optimization.

MILP models are typically solved using exact algorithms, such as Branch-and-Bound and Cutting Plane methods, or heuristic and metaheuristic techniques, including Tabu Search, Simulated Annealing, and Genetic Algorithms, particularly when problem size leads to computational intractability.

For CCUS network design, MILP-based optimization frameworks support integrated system planning, balancing cost minimization with CO_2 emission mitigation objectives (Lee et al., 2014; Middleton and Bielicki, 2009; Zhang et al., 2020b). These methods provide flexibility to evaluate multiple scenarios and inform strategic investment decisions for sustainable CCUS deployment.

4. Method

This study adopts a Systematic Literature Review (SLR) to comprehensively analyze advancements in mathematical modeling for CCS/CCUS supply chain network design. An SLR is a structured and transparent approach that synthesizes existing research using predefined and reproducible procedures, thereby minimizing selection bias and ensuring methodological rigor (Higgins and Thomas, 2024). This review was conducted and reported in accordance with the Preferred Reporting

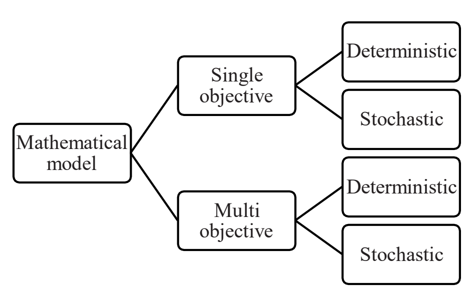

Fig. 5. Schematic of mathematical model.

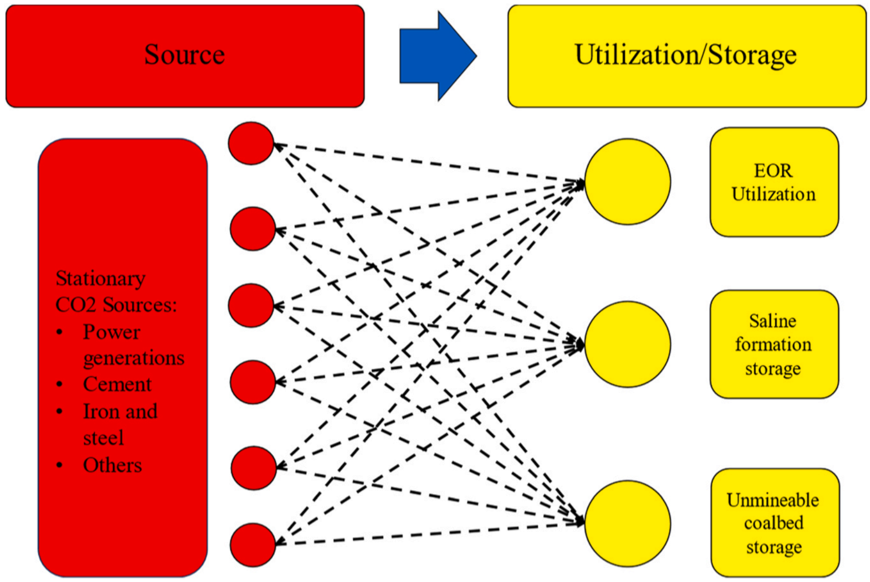

Fig. 6. Source–Sink matching network diagram (edited from ref: (Jiao et al#, 2024)).

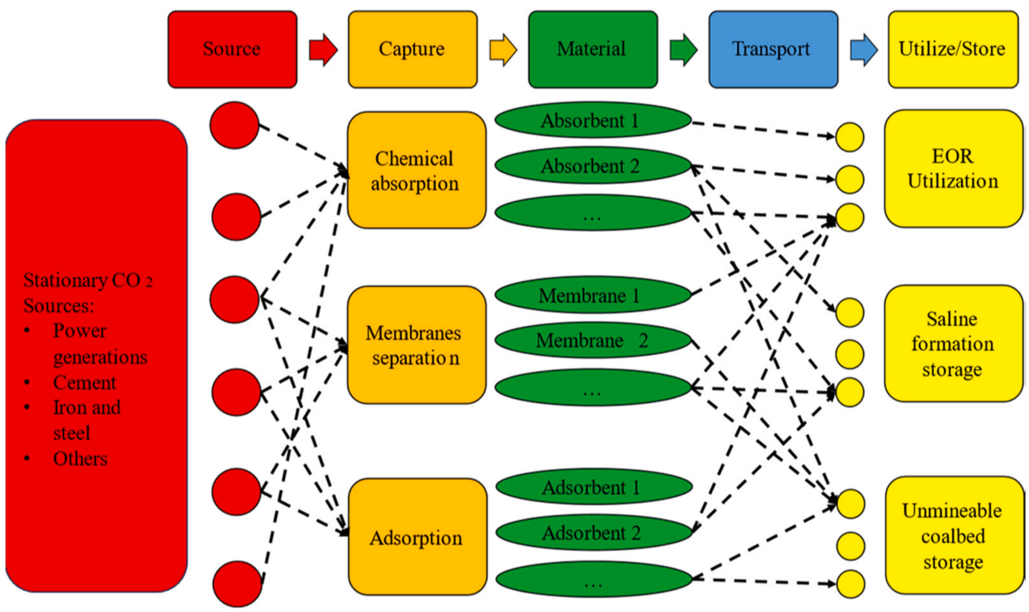

Fig. 7. Typical of multi-echelon CCUS supply chain network illustrating source nodes, capture technologies, and utilization or storage sites (Hasan et al#, 2015).

Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement, which provides updated guidance to enhance transparency, completeness, and consistency in systematic review reporting (Page et al., 2021).

In line with established SLR practices, the review process involved: (i) defining explicit research questions, (ii) establishing a transparent and reproducible search strategy, (iii) applying clearly defined inclusion and exclusion criteria, (iv) conducting systematic screening and eligibility assessments, and (v) synthesizing findings to support evidence-based conclusions and future research directions (Higgins and Thomas, 2024; Mengist and Soromessa, 2020).

4.1. Research questions

This review focuses on recent developments in mathematical modeling approaches for CCUS supply chain network design. The study is guided by the following research questions:

a. How has the scope of modeling evolved?
b. What are the main mathematical modeling approaches used in CCUS supply chains?
c. What are the main objectives considered in CCUS supply chain models?
d. How are case studies of these models solved?

4.2. Search strategy

A comprehensive literature search was conducted in the Web of Science and Scopus databases on July 28, 2025. These databases were selected due to their extensive coverage of peer-reviewed journals in engineering, energy systems, and optimization research. The search was applied to titles, abstracts, and author keywords using the following Boolean query:

("Carbon Capture and Storage" AND "Mathematical Model") OR ("Carbon Capture, Utilization, and Storage" AND "Mathematical Model")

OR ("Carbon Capture and Storage" AND "Mixed-Integer Linear Programming") OR ("Carbon Capture, Utilization, and Storage" AND "Mixed-Integer Linear Programming")

The search strategy was designed to capture studies addressing CCUS supply chain modeling using mathematical programming, with particular emphasis on *mixed-integer linear programming* (MILP) and related optimization techniques.

4.3. Study selection process

The article selection process followed the PRISMA 2020 framework and consisted of four main stages: identification, screening, eligibility, and inclusion. A detailed summary of this process is presented in Table 1.

4.3.1. Identification

The initial database search yielded a total of 476 records, comprising 155 articles from Web of Science and 321 articles from Scopus. All retrieved records were exported in BibTeX format and merged into a single database.

4.3.2. Screening

During the screening stage, records were filtered based on predefined inclusion criteria applied directly within the databases. The screening criteria included:

- publication year between 2010 and 2025,
- document type restricted to peer-reviewed research articles,
- publication language limited to English.

After applying these filters, duplicate records were removed, resulting in 228 unique articles for further assessment. A total of 126 duplicate records were excluded at this stage.

4.3.3. Eligibility

Eligibility assessment was conducted in two sequential steps to ensure thematic relevance and methodological alignment.

First, titles and abstracts were reviewed to identify studies addressing CCUS or CCS topics using mathematical programming or MILP-based approaches. Based on this assessment, 96 articles were retained, while 132 records were excluded due to insufficient topical relevance.

Second, full-text screening was performed to evaluate the alignment of the remaining studies with CCUS supply chain network design and the application of mathematical programming methods. This step resulted in the exclusion of 44 articles, yielding a final set of 52 eligible studies.

4.4. Inclusion and exclusion criteria

Studies were included in the review if they met all of the following criteria:

- published in peer-reviewed journals between 2010 and 2025,
- written in English,
- focused on CCS or CCUS supply chain network design,
- employed MILP or other mathematical programming approaches.

Studies were excluded if they:

- were conference papers, reports, or non-peer-reviewed publications,
- did not address CCS or CCUS-related supply chain modeling,
- focused solely on technical capture processes without supply chain or network design considerations,
- lacked sufficient methodological detail.

4.5. Quality verification and data management

To ensure data accuracy and consistency, all records were managed using the Rayyan AI platform, which facilitated duplicate removal, title and abstract screening, and record classification. The selection process was performed by a single person using Rayyan AI, allowing efficient organization and filtering of the retrieved data. Incomplete records, non-English publications, and non–peer-reviewed studies were excluded during this stage.

All authors subsequently reviewed and validated the screening and eligibility results to ensure consistency and minimize subjective bias. Abstracts and full-text articles were carefully examined to confirm their relevance, methodological rigor, and alignment with the study objectives. The final dataset of 52 articles was collaboratively validated to maintain research integrity and analytical reliability.

The complete flow of article identification, screening, eligibility assessment, and inclusion is summarized in Table 1, which adheres to the PRISMA 2020 reporting structure and provides transparency throughout the selection process.

5. Results and discussion

5.1. Article distribution

Fig. 8 shows the publication trend related to the use of MILP in the CCUS supply chain. Overall, there has been an increase in the publication trend from 2012 to 2025, with the highest number of publications occurring in 2020. Fig. 9 displays the names of the journals related to these publications. The Journal of Cleaner Production, Computers and Chemical Engineering Journal, and Applied Energy are the top three journals publishing articles on the use of MILP in the CCUS supply chain. While, Table 2 provides an overview of the first author's last name and the frequency of their publications included in the systematic review.

Table 1
The flow diagram for the database search of publications for systematic reviews (inspired from ref: (Mengist and Soromessa, 2020; Rakhshan et al., 2020).

Step	Databases Searching string and searching terms	No of articles (results)	
		Web of Science	Scopus
Identification	Using the query and keywords as described above	155	321
Screening	Published year (2010–2025)*	155	307
	Only peer-reviewed research and review articles	131	232
	Language: English	129	225
	Total articles	354	
	Remove duplicates	228 (Excluded = 126)	
Eligibility	Domain: Title and abstract reading	96 (Excluded = 132)	
	Approach: Topic Similarity with CCS/CCUS & MILP/Mathematical Programming Methods		
	Domain: Main body reading	52 (Excluded = 44)	
	Approach: Approach: Topic similarity between SCND in CCS/CCUS and the use of MILP/Mathematical Programming Methods		
Included	Papers included to be reviewed	52	

* as of the database search date, July 28, 2025

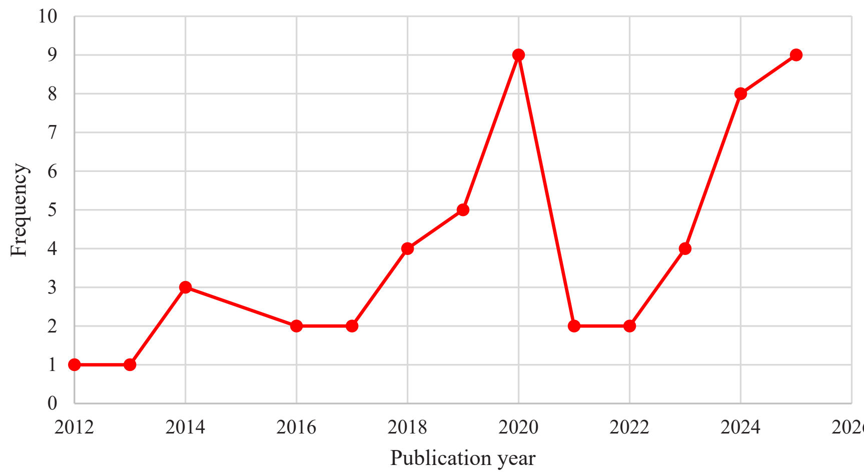

Fig. 8. Trends of publications over the years.

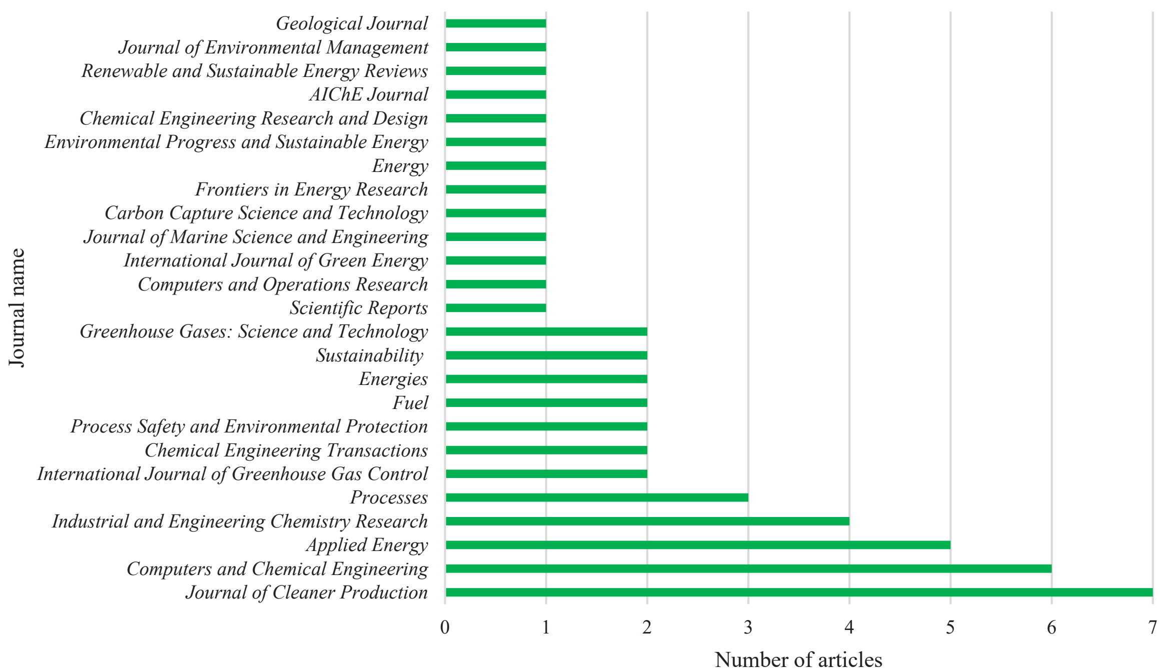

Fig. 9. The distribution of articles based on the source of publication.

Table 2
First author's last name and frequency from the systematic review.

First author's last name	Frequency	References.
d'Amore	8	(D'Amore et al., 2018; d'Amore et al, 2019; D'Amore et al., 2019; D'Amore and Bezzo, 2020; D'Amore et al., 2020, 2021a, 2021b, 2024)
Zhang	7	(Zhang et al., 2018a, 2020a, 2020b, 2017, 2018b, 2019a, 2019b)
Leonzio	5	(Leonzio et al., 2019a, 2019b, 2020a; Leonzio and Zondervan, 2020; Leonzio et al., 2020b)
Tapia	3	(Frederick et al., 2016a; Tapia and Tan, 2014; Tapia et al., 2016a)
Zhou	3	(Zhou et al., 2023, 2024, 2025)

5.2. Solver and software distribution

A software interface is essential for performing optimization, enabling the connection to solvers that resolve optimization problems. MILP solvers are used to solve optimization problems involving both integer and continuous variables through methods such as branch and bound, cutting planes, and heuristics (Kumar and Mageshvaran, 2020). Fig. 10 illustrates the distribution of software and solver usage in mathematical modeling for CCUS supply chains. Among these, GAMS and the CPLEX solver are the most commonly used.

5.3. Modelling scope

Fig. 11 shows the locations of case studies on the CCUS supply chain, with China and Europe being the top regions for these publications. Table 3 provides a comprehensive summary of various recent studies related to the CCUS supply chain, including case study locations, modeling tools, and modeling scopes.

The scope of mathematical modeling in the CCUS supply chain is illustrated in Fig. 12. Essentially, CCS/CCUS modeling covers fundamental aspects such as the CO_2 source/capture plant, the CO_2 sink (storage or utilization), and the volume of CO_2 flow. Further

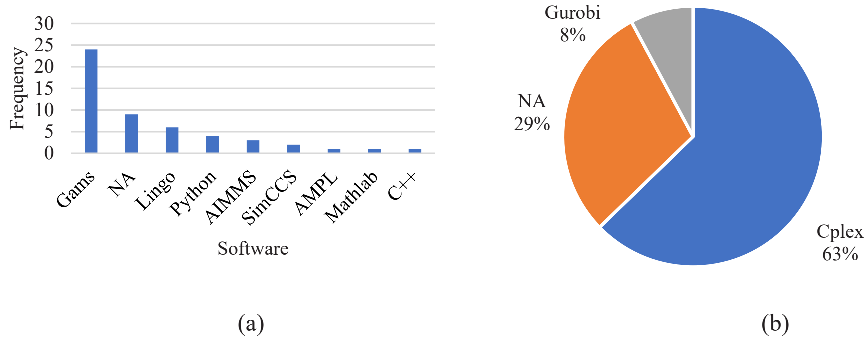

Fig. 10. Distribution of articles based on: (a) modeling software and (b) optimization solver used.

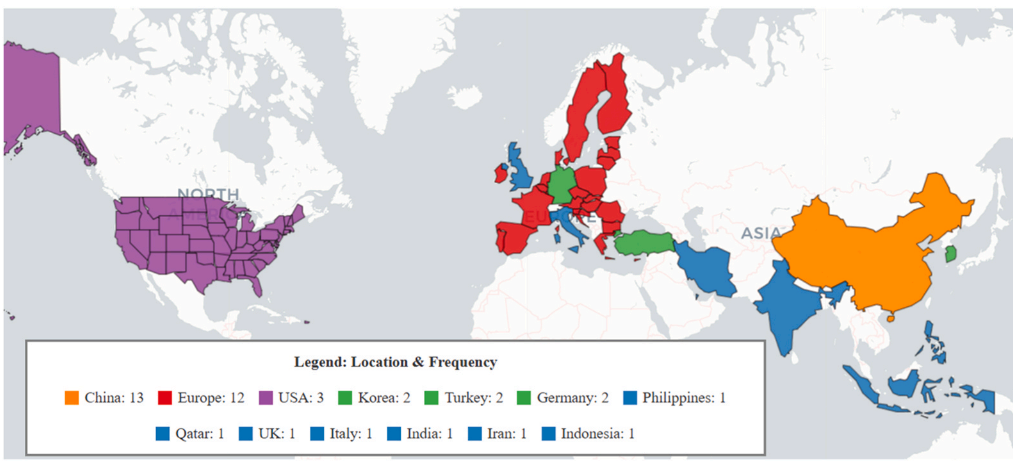

Fig. 11. Geographic distribution of CCUS supply chain case study locations and article frequency in the SLR.

development involves the optimization of the transportation network. Other aspects such as CO_2 price/policy, environmental impact, social considerations, resilience, and risk are still rarely addressed.

5.4. Classification of mathematical models

5.4.1. Single objective function

5.4.1.1. Deterministic. Table 4 presents the characteristics of CCS/CCUS modeling structures with single-objective deterministic formulations. Several studies have employed exact algorithms to achieve globally optimal solutions, particularly for small- to medium-sized datasets. Tan et al. applied an exact algorithm to a simple deterministic MILP model (Tan et al., 2013), while Tapia et al. used exact algorithms with discrete-time optimization to schedule EOR operating lifetimes (Frederick et al., 2016b). Zhang et al. applied exact algorithms with Nash game theory for fair CCS infrastructure design (Zhang et al., 2017) and compared source-sink matching with integrated CCUS supply chain superstructures (Zhang et al., 2018a). D’Amore proposes a multi-echelon, spatially-explicit MILP model solved with an exact optimization method to minimize costs while considering societal risk constraints, achieving efficient solutions on a standard laptop; for larger-scale problems, heuristic or advanced exact algorithms may be required (D’Amore et al., 2018). Leonzio & Zondervan modeled multi-echelon CCUS supply chains in Germany using exact algorithms (Leonzio et al., 2019b), and Ayyad et al. applied goal programming with exact optimization to balance costs, CO_2 exchange, and storage utilization (Ayyad et al., 2024).

Shikha et al. employed exact MILP with decomposition and rolling/shrinking horizons to solve large-scale milk-run scheduling problems, maintaining optimality for each subproblem (Shikha et al., 2025), while Agrali et al. and Esmaeili transformed nonlinearities into MILPs and solved them exactly using linearization (Agrali et al., 2018) and Stackelberg-diagonalization techniques (Esmaeili, 2020). Additional studies using exact MILP/optimization include (Zhang et al., 2018a; Frederick et al., 2016a; Lee et al., 2014; Zhang et al., 2020b; D’Amore et al., 2019; D’Amore and Bezzo, 2020; D’Amore et al., 2021a; Leonzio et al., 2019b, 2020a; Leonzio and Zondervan, 2020; Zhou et al., 2023, 2024; Patange et al., 2022; Wu et al., 2022; Ma et al., 2023; Chen et al., 2024; Ayyad et al., 2024; Shikha et al., 2025; Guisso et al., 2025; Liu et al., 2025; Esmaeili, 2020; D’Amore et al., 2021c).

In contrast, some studies adopted heuristic or hybrid approaches to handle larger or more complex CCS/CCUS problems where exact methods become computationally expensive. Jin et al. used a central limit theorem-based heuristic to identify candidate intermediate storage sites (Jin et al., 2017). Tao et al. implemented a hybrid two-stage method, solving site selection for small- to medium-scale instances with exact MIP (Gurobi) and ship assignment and routing for large-scale problems using a Genetic Algorithm (Tao et al., 2025). Homsi et al. applied a slope scaling heuristic with iterative approximation and dynamic programming for multi-period, geographically explicit pipeline-based CCS planning, generating high-quality solutions that often outperform Cplex but do not guarantee global optimality (Homsi et al., 2025).

5.4.1.2. Non-deterministic. Table 5 summarizes the characteristics of CCS/CCUS modeling frameworks with single-objective non-deterministic formulations. Tapia develops a novel fuzzy-MILP model using a fuzzy optimization approach and an exact algorithm to optimize source-sink matching in multi-period CCS systems, balancing technical risks

Table 3
CCS/CCUS modelling scope.

Ref.	Case study	Modeling Tools		Modeling scope									
		Software	Solver	CO_2 flow	Carbon Capture Plant	CO_2 Demand (Storage)	CO_2 Demand (Utilization)	Transp.	Env. Effect	Social	CO_2 Price/ Policy	Resilience	Risk
(Lee et al., 2012)	Korea	Gams	Cplex	v	v	v	v	v	v				
(Tan et al., 2012)	NA	Lingo	NA	v	v	v							
(Tapia and Tan, 2014)	Philippines	Lingo	NA	v	v	v							
(Lee et al., 2014)	NA	Gams	Cplex	v	v	v							
(He et al., 2014)	NA	Gams	Cplex	v	v	v							
(Tapia et al., 2016b)	NA	Lingo	NA	v	v	v	v	v			v		
(Frederick et al., 2016b)	NA	Lingo	NA	v	v		v	v			v		
(Zhang et al., 2017)	Qatar	Gams	Cplex	v	v	v	v	v			v		
(Jin et al., 2017)	Korea	NA	NA	v	v	v		v					
(Agrali et al., 2018)	Turkey	Gams	Cplex	v	v	v	v	v					
(D'Amore et al., 2018)	Europe	Gams	Cplex	v	v	v		v		v			v
(Zhang et al., 2018b)	NA	NA	NA	v	v	v							v
(Zhang et al., 2018a)	China	Gams	Cplex	v	v	v	v	v			v		
(Leonzio et al., 2019a)	Europe	NA	Cplex	v	v	v	v	v					
(Zhang et al., 2019a)	China	Gams	Cplex	v	v	v		v			v		v
(d'Amore et al, 2019)	Europe	Gams	Cplex	v	v	v		v					v
(Zhang et al., 2019b)	China	Gams	Cplex	v	v	v	v	v	v		v		
(Leonzio et al., 2019b)	Germany	AIMMS	Cplex	v	v	v	v	v					
(D'Amore et al., 2019)	Europe	Gams	Cplex	v	v	v	v	v			v		
(Leonzio et al., 2020a)	UK	AIMMS	Cplex	v	v	v	v	v			v		
(Aliabadi, 2020)	Turkey	NA	Cplex	v	v	v	v	v					
(Leonzio and Zondervan, 2020)	Italy	NA	Cplex	v	v	v	v	v			v		
(Zhang et al., 2020b)	China	Gams	Cplex	v	v	v	v	v			v		
(Leonzio et al., 2020b)	Europe	NA	Cplex	v	v	v	v	v			v		
(D'Amore and Bezzo, 2020)	Europe	Gams	Cplex	v	v	v	v	v			v		
(Zhang et al., 2020a)	China	Gams	Cplex	v	v	v	v	v	v		v		
(D'Amore et al., 2020)	Europe	Gams	Cplex	v	v	v		v		v			v
(D'Amore et al., 2021a)	Europe	NA	NA	v	v	v		v					
(D'Amore et al., 2021b)	Europe	Gams	Cplex	v	v	v		v					
(Patange et al., 2022)	India	AMPL	Cplex	v	v	v	v	v			v		
(Wu et al., 2022)	China	Lingo	NA	v	v	v	v	v			v		

(continued on next page)

Table 3 (*continued*)

Ref.	Case study	Modeling Tools		Modeling scope									
		Software	Solver	CO_2 flow	Carbon Capture Plant	CO_2 Demand (Storage)	CO_2 Demand (Utilization)	Transp.	Env. Effect	Social	CO_2 Price/ Policy	Resilience	Risk
(Ma et al., 2023)	USA	SimCCS	Cplex	v	v	v		v					
(Zhou et al., 2023)	China	Gams	Cplex	v	v	v	v	v			v		
(Abdoli et al., 2023)	NA	Gams	Cplex	v	v*	v	v	v			v		
(Nie et al., 2023)	China	Python	Gurobi	v	v	v	v	v	v		v		
(Chen et al., 2024)	China	Mathlab	Gurobi	v	v	v		v					
(Shirazaki et al., 2024)	Iran	Gams	Cplex	v	v	v	v	v			v		
(Derakhti et al., 2024)	Europe	Gams	Cplex	v	v	v			v	v	v		
(Ayyad et al., 2024)	Indonesia	Lingo	NA	v	v	v							
(D'Amore et al., 2024)	Europe	NA	NA	v	v	v		v					
(Zhou et al., 2024)	China	Gams	NA	v	v	v	v	v			v		
(Nguyen et al., 2024)	Germany	AIMMS	Cplex	v	v	v	v	v	v		v		
(Lei et al., 2024)	China	C+ +	Cplex	v	v	v	v	v			v	v	
(Tao et al., 2025)	Europe	Python	Gurobi	v	v	v		v**					
(Shikha et al., 2025)	NA	Gams	Gurobi	v	v	v		v**					
(Guisso et al., 2025)	NA	Gams	Cplex	v	v	v		v**					
(Liu et al., 2025)	China	Gams	NA	v	v	v	v	v			v		
(Homsi et al., 2025)	USA	Python	Cplex	v	v	v		v					
(Gu et al., 2025)	NA	NA	NA	v	v	v		v					
(Olson and Yaw, 2025)	USA	SimCCS	NA	v	v	v		v					
(Zhou et al., 2025)	China	Python	NA	v	v	v	v	v			v		
(Wu et al., 2025)	NA	NA	NA	v	v	v	v	v			v		

NA: Not Available information in the articels
*Using single CO_2 source
**Focusing in the maritime/ship transportaion

and CO2 emissions reduction under uncertainties in storage site properties (Tapia and Tan, 2014). Leonzio applies an exact algorithm to solve a CCUS optimization problem, while incorporating Monte Carlo simulation to account for uncertainties in demand and selling prices through a probabilistic approach (Leonzio et al., 2019a). The use of exact algorithms has also been employed in several other studies, including (D'Amore et al., 2021a; Leonzio et al., 2020b; Abdoli et al., 2023; Nie et al., 2023; Shirazaki et al., 2024; Lei et al., 2024; Olson and Yaw, 2025; Wu et al., 2025; Yi-Jun et al., 2014).

5.4.2. *Multi objective function*

5.4.2.1. Deterministic. Lee et al.'s multi-objective model for optimal CCS infrastructure planning in Korea was solved using the ε-constraint method, achieving quick solutions with low optimality gaps through exact optimization. The Pareto optimal solutions balance economic performance and environmental damage, with impacts assessed using the Eco-indicator 99 method based on Life Cycle Assessment (LCA) principles (Lee et al., 2012). Zhang et al. develop a multi-objective MILP framework for the CCUS supply chain in Northeast China, optimizing total annual cost and life cycle GHG emissions over a 20-year horizon. Using the ε-constraint method as an exact algorithm, the Pareto front illustrates the trade-off between cost and emissions (Zhang et al., 2019b). Similarly, Zhang et al.'s multi-objective MILP model aims to minimize both total cost and environmental impact, measured by Global Warming Potential (GWP) based on LCA principles, using the ε-constraint method and an exact algorithm to optimize the CCUS supply chain in Northeast China (Zhang et al., 2020a). D'Amore et al. develop a multi-objective MILP model for European CCS supply chains, optimizing costs and social acceptance using the ε-constraint method, an exact algorithm for Pareto-based bi-objective optimization. The results demonstrate the trade-off between economic and social objectives, with an intermediate solution balancing both and highlighting the conflict between costs and risk perception (D'Amore et al., 2020).

In Derakhti and Santibanez's study, a bi-objective optimization approach was solved using the Pareto method for multi-objective optimization and an exact algorithm (Derakhti et al., 2024). Nguyen et al. applied the ε-constraint technique and an exact algorithm to solve a bi-objective MILP model for optimizing the CCUS supply chain in Germany, generating a Pareto front that shows the trade-off between economic efficiency and GHG reduction (Nguyen et al., 2024).

Most studies employed multi-objective MILP models solved using the

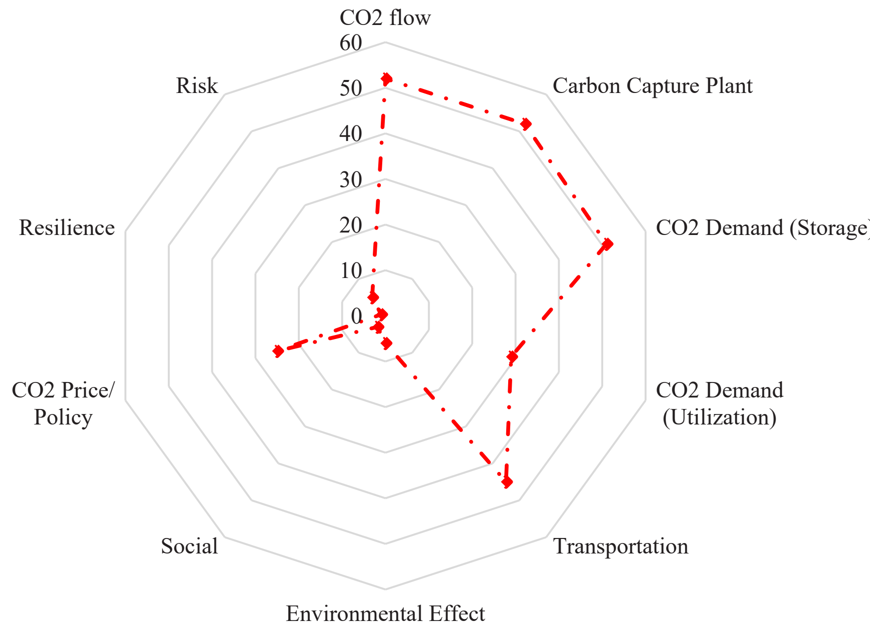

Fig. 12. Radar diagram of modeling scope across CCS/CCUS supply chain studies.

ε-constraint method, an exact algorithm that efficiently generates Pareto-optimal solutions, revealing trade-offs between cost, emissions, and other criteria. This approach was widely applied across regional CCUS planning studies in Asia and Europe. In contrast, Gu et al. used a genetic algorithm for source–sink matching due to the NP-complete nature of the problem, prioritizing computational speed over exact Pareto solutions (Gu et al., 2025). The detailed model structure is provided in Table 6.

5.4.2.2. Non deterministic. Zhang et al. develop a two-stage stochastic MILP model using a finite-scenario approach to optimize carbon capture retrofitting and CO_2 source–sink matching under uncertainty. The model incorporates a probabilistic financial risk metric and is solved using stochastic programming techniques (Zhang et al., 2018b). In a subsequent study, Zhang et al. propose a risk management optimization framework for CCS deployment in Northeast China, comparing a risk-neutral two-stage stochastic MILP with risk-averse multi-objective models that account for carbon tax uncertainty. First-stage variables represent infrastructure design decisions, while second-stage variables capture operational decisions. Using the ε-constraint method, the model generates Pareto-optimal solutions that reveal trade-offs between cost and risk, enhancing decision-making compared to risk-neutral approaches (Zhang et al., 2019a). The detailed model structure is summarized in Table 7.

Fig. 13 shows the distribution of CCUS supply chain model structures. It can be seen that the majority of models (83%) have a single-objective function, while the remaining 17% are multi-objective. Models with a two-echelon structure account for 44%, and the remaining 56% are multi-echelon. Deterministic models make up 75%, while the remaining 25% are non-deterministic or incorporate uncertainty.

5.5. Modelling uncertainity in CCS/CCUS

SCND faces uncertainties in costs, demand, supply, and rare high-impact events. These uncertainties—aleatory (probabilistic) or epistemic (knowledge-based)—create gaps between known and needed information, and SCND aims to maintain optimal performance across scenarios (Govindan et al., 2017; Acar et al., 2021). The main approaches to optimization under uncertainty include (Sahinidis, 2004):

1. Stochastic Programming: This involves recourse models, where decisions are made in two stages — the first stage involves making decisions before uncertainty is realized, and the second stage addresses the changes in decisions once uncertainty is known. It also includes robust stochastic programming, which aims to optimize for the worst-case scenario, and probabilistic models (chance-constrained programming), which focus on the reliability and probability of satisfying constraints under uncertainty.
2. Fuzzy Programming: This approach includes flexible programming, which deals with uncertainties in the right-hand side of constraints, and possibilistic programming, which handles uncertainty in both objective functions and constraint coefficients. Uncertainty is modeled using fuzzy numbers and constraints are treated as fuzzy sets, allowing some degree of violation of constraints.
3. Stochastic Dynamic Programming: This method is used for problems where decisions are made over multiple stages and the future state depends on current decisions, with uncertainty affecting outcomes at each stage.

Uncertainty in CCS systems arises from multiple sources that affect their overall reliability and efficiency. Broadly, these uncertainties can be categorized into two main groups: (1) uncertainties related to CO_2 sources, such as variations in emission rates and the operational lifespan of emission facilities, and (2) uncertainties associated with CO_2 sinks, including limitations in injection rates and storage capacities (Diamante et al., 2013; Middleton et al., 2012). Within the storage subsystem, key uncertainties stem from reservoir characteristics such as permeability, porosity, and capacity, which directly influence CO_2 storage performance (Keating et al., 2011; Lee et al., 2017). These geological uncertainties often lead to variable injectivity, uneven plume migration, and potential requirements for additional wells, thereby increasing costs (Diamante et al., 2013). Furthermore, storage capacity uncertainties arise from limited data accuracy and inconsistent efficiency factors, which can result in unreliable capacity estimates and lost economic opportunities (Suzuki et al., 2013; Bradshaw et al., 2007). Beyond geological factors, supply chain-level uncertainties—including fluctuations in CO_2 generation, variability in construction and operational costs, evolving capture technologies, and external influences such as policy, market forces, and engineering performance—further complicate CCS system planning and optimization (Zhang et al., 2014; Malaeb

Table 4
Summary of CCS/CCUS Optimization Studies Focused on Single-Objective and Deterministic Approaches.

Reference	Objective function type			Network structure		Uncertainty Classification	
	Objective function	SO	MO	Two-echelon	Multi-echelon	Det.	Non Det.
(Tan et al., 2013)	Maximize the total amount of CO_2 captured and stored	v		v		v	
(Lee et al., 2014)	Maximize net CO_2 reduction over time, considering capture and energy (power) penalties.	v		v		v	
(Tapia et al., 2016b)	Maximize profit from oil revenue and carbon credits minus total system cost.	v		v		v	
(Frederick et al., 2016b)	Maximize profit from oil revenue and CO_2 credits minus infrastructure costs	v		v		v	
(Zhang et al., 2017)	Minimizes total cost, equalizes savings, and ensures fairness using Nash bargaining.	v			v	v	
(Jin et al., 2017)	Minimize total CCS project cost, including capital costs for CO_2 capture facilities and infrastructure and operating costs for CO_2 pipelines.	v		v		v	
(D'Amore et al., 2018)	Minimize total CCS costs—capture, transport, and sequestration—while ensuring societal risk stays below acceptable regional limits through appropriate risk mitigation.	v			v	v	
(Agrali et al., 2018)	Minimize net present cost of CCS, including capture, storage, transport, emissions trading, and carbon utilization.	v			v	v	
(Zhang et al., 2018a)	Minimize the annualized net cost of the CCUS network, defined as the difference between total annual cost and total annual revenue	v			v	v	
(Leonzio et al., 2019b)	Minimize the total cost of the CCUS supply chain, including CO_2 capture and compression, transport, storage, and the production and transport of methanol, methane, steam, and hydrogen.	v			v	v	
(D'Amore et al., 2019)	Minimizes the total European CCS network cost at each iteration, covering installation and operation expenses across capture, transport, and storage stages.	v			v	v	
(Leonzio et al., 2020a)	Minimizes total CCUS supply chain cost, including CO_2 capture, transport, storage, and production of CO?-based products.	v			v	v	
(Aliabadi, 2020)	Minimizes the total system cost, including investment, fixed and variable operating costs, storage, and transportation, while subtracting revenues from CO_2 utilization and credit trading—effectively maximizing the net present value.	v		v		v	
(Leonzio and Zondervan, 2020)	Minimizes total supply chain cost, including CO_2 capture, transport, storage, and methane production	v			v	v	
(Zhang et al., 2020b)	Minimizes the annualized net cost of the CCUS supply chain by optimizing costs across capture, transport, storage, and utilization, minus revenues from CO_2 product sales.	v			v	v	
(D'Amore and Bezzo, 2020)	Minimize the total cost of the European CCUS network, combining capture, transport, and storage costs while subtracting profits from CO_2 utilization.	v			v	v	
(D'Amore et al., 2021a)	Minimizes total annual CCS system cost, comprising capture, transport, and sequestration costs.	v			v	v	
(D'Amore et al., 2021c)	Minimizes the total cost of the CCS supply chain, which includes the costs of CO_2 capture, transportation, and sequestration.	v			v	v	
(Patange et al., 2022)	Minimizes the total annualized cost of a biomass–CO_2–EOR supply chain, including biomass processing, bioenergy production, CO_2 capture, transport, and oil recovery, offset by revenues from bioethanol and electricity sales.	v		v		v	
(Wu et al., 2022)	Minimize the discounted net cost of the full CCUS chain, combining capture, transport, utilization, and storage costs minus CO_2 utilization revenues.	v			v	v	
(Ma et al., 2023)	Minimizes total CCS costs, covering capture, transport, and storage over the project's time horizon.	v		v		v	
(Zhou et al., 2023)	Minimizes annual net cost by summing capture, transport, and storage costs, then subtracting utilization revenue.	v		v		v	
(Chen et al., 2024)	Minimizes total CCUS costs for capture, transport, and storage, including fixed, O&M, variable, and injection well costs over the project lifetime.	v		v		v	
(Ayyad et al., 2024)	The goal programming model optimizes the CO_2 network by balancing cost, CO? exchange, and storage use, minimizing the gap between each goal's target and actual outcome based on set priorities.	v			v	v	
(D'Amore et al., 2024)	Minimises the total annual cost of a ship-based CO_2 transport chain, including all stages from liquefaction to unloading, based on capital and operating expenses and optimal logistics between capture and sequestration points.	v			v*	v	
(Zhou et al., 2024)	Minimise the total annualised investment and operational costs for carbon reduction, including flue gas dehydration, CO_2 capture and compression, pipeline construction and operation, CO_2 storage, carbon tax payments, and carbon trading.	v			v	v	
(Tao et al., 2025)	Minimise total costs, including CO_2 storage construction, ship chartering, transport, and emission penalties.	v		v		v	
(Shikha et al., 2025)	Maximise net profit by increasing CO_2 shipped from emitters to the terminal while accounting for penalties, operating expenses, and vessel fixed costs.	v		v		v	
(Guisso et al., 2025)	Maximise CCS project profit over a set time horizon by increasing cryogenic CO_2 transport to the terminal, minimising operating costs and vessel usage, and penalising CO_2 venting at emitters.	v		v		v	
(Liu et al., 2025)	Minimise annualised costs for CO_2 capture, transport, storage, and carbon taxes.	v		v		v	
(Homsi et al., 2025)	Minimises total discounted CCS costs for capture, transport, and storage	v		v		v	
(Zhou et al., 2025)	Minimize yearly CCUS costs for capture, transport, storage, carbon tax payments, and carbon trading.	v			v	v	

et al., 2019).

Table 8 summarizes key studies addressing uncertainty in CCS and CCUS supply chain and network optimization. Across the literature, uncertainties typically stem from both CO_2 sources—such as emission variability, lifespan of capture facilities, and operational constraints—and CO_2 sinks, particularly in the form of geological uncertainty regarding storage capacity, permeability, and injectivity. Broader system-level uncertainties, including fluctuations in investment cost,

Table 5
Summary of CCS/CCUS Optimization Studies Focused on Single-Objective and Non-Deterministic Approaches.

Reference	Objective function type			Network structure		Uncertainty Classification	
	Objective function	SO	MO	Two-echelon	Multi-echelon	Det.	Non Det.
(Tapia and Tan, 2014)	Maximize satisfaction level considering CO_2 reduction and technical risk under uncertainty.	v		v			v
(He et al., 2014)	Maximize total CO_2 stored.	v		v			v
(Leonzio et al., 2019a)	Minimize total cost, including CO_2 capture, transport, storage, and various utilization product costs.	v			v		v
(d'Amore et al, 2019)	Minimizes total CCS cost by including capture, transport, sequestration, and quantified risk from storage uncertainty.	v			v		v
(Leonzio et al., 2020b)	Minimizes the expected total cost of the CCUS supply chain, including capture, transport, storage, and CO_2-based product production costs	v			v		v
(Abdoli et al., 2023)	Maximizes total profit from carbon credits and EOR revenues, minus the fixed and variable costs of CCS network design, CO_2 transfer, and pipeline infrastructure	v		v			v
(Nie et al., 2023)	Minimizes the total CCUS supply chain cost, including costs of capture, transport, utilization, storage, and associated operations	v			v		v
(Shirazaki et al., 2024)	Minimizes net annualized CCUS supply chain cost by reducing expenses for capture, transport, and storage while accounting for revenues from carbon utilization, using scenario-based programming to handle uncertainties.	v			v		v
(Lei et al., 2024)	Minimise total supply chain costs, including investment, operating, and transit costs	v			v		v
(Olson and Yaw, 2025)	Minimizes total capture, transport, and storage costs across the network.	v		v			v
(Wu et al., 2025)	Minimizes total CCUS system costs, covering capture, transport, storage, recovery, and variable operation costs	v		v			v

Table 6
Summary of CCS/CCUS Optimization Studies Focused on Multi-Objective and Deterministic Approaches.

Reference	Objective function type			Network structure		Uncertainty Classification	
	Obj. function	SO	MO	Two-echelon	Multi-echelon	Det.	Non Det.
(Lee et al., 2012)	Minimizes total annual cost and environmental impact (Eco-indicator 99)		v	v		v	
(Zhang et al., 2019b)	Minimizes the annualized net cost of the CCUS supply chain—defined as total annual cost minus CO_2 utilization revenue—while also minimizing life cycle greenhouse gas emissions		v		v	v	
(Zhang et al., 2020a)	Minimizes both total annual cost and environmental impact of the CCUS supply chain.		v		v	v	
(D'Amore et al., 2020)	Minimizes both total cost (capture, transport, and storage) and public risk perception associated with CCS infrastructure.		v		v	v	
(Derakhti et al., 2024)	The model first minimizes CCS supply chain cost, then applies a bi-objective approach to also reduce social risk perception—based on cultural factors and nearby population—to enhance social acceptance.		v		v	v	
(Nguyen et al., 2024)	Maximise both the CCUS system's annual profit and life cycle GHG reduction.		v		v	v	
(Gu et al., 2025)	The objective functions aim to maximize storage capacity and minimize transportation costs		v	v		v	

Table 7
Summary of CCS/CCUS optimization studies focused on multi-objective and non-deterministic approaches.

Reference	Objective function type			Network structure		Uncertainty Classification	
	Objective function	SO	MO	Two-echelon	Multi-echelon	Det.	Non Det.
(Zhang et al., 2018b)	Maximizes net CO_2 storage while minimizing financial risk using a weighted probabilistic approach.		v	v			v
(Zhang et al., 2019a)	Minimizes expected total cost, combining fixed capital costs for capture, storage, and pipelines with expected operating and maintenance costs across all scenarios.		v		v		v

energy penalties, and carbon policy parameters, also play a significant role in determining optimal infrastructure design and financial feasibility.

Early works, such as Tapia and Tan, applied fuzzy FMILP to capture imprecise and linguistic uncertainties in reservoir characteristics, introducing the *max–min aggregation* principle to balance reduction goals against technical feasibility. Subsequent studies transitioned toward stochastic programming, which represents uncertain parameters through probability distributions (Tapia and Tan, 2014).

Several studies apply stochastic MILP to manage uncertainty in CCS/CCUS systems. He et al. incorporate worst-case and robust stochastic formulations to address variability in CO_2 source lifetimes, storage capacities, and compensatory power (Yi-Jun et al., 2014). Leonzio et al. introduced a stochastic MILP coupled with Monte Carlo Simulation to capture market-driven uncertainty in CO_2-based product prices through probabilistic sampling, enhancing model realism and cost estimation

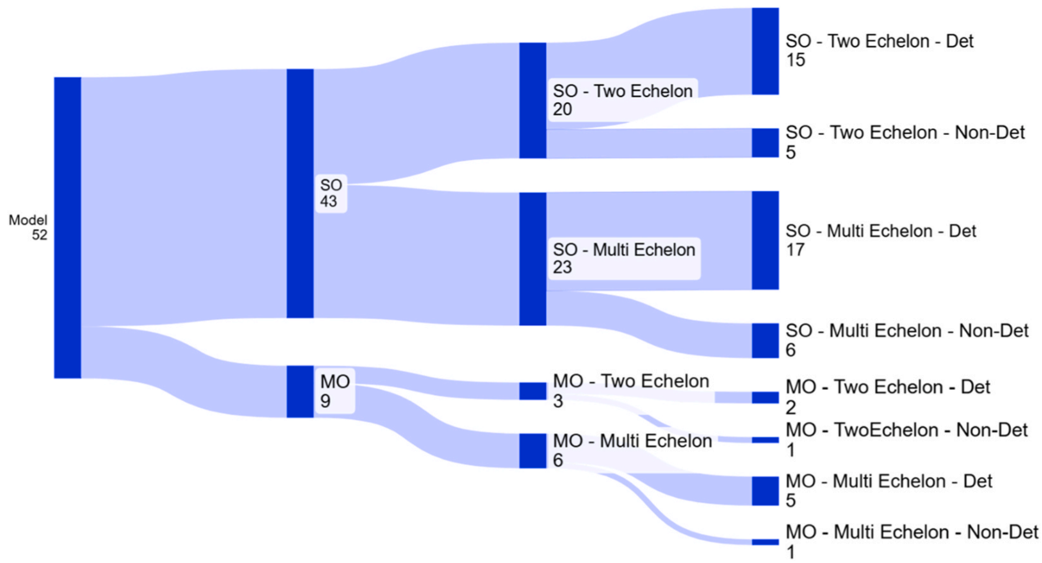

Fig. 13. Sankey diagram in mathematical modeling of CO_2 supply chain in CCUS.

Table 8
Summary of uncertainty sources and corresponding modeling approaches applied in CCS/CCUS system optimization studies.

Reference	Uncertainty Sources	Modeling Approach
(Tapia and Tan, 2014)	CO_2 storage capacity and injectivity	Fuzzy MILP using *max–min* aggregation
(Yi-Jun et al., 2014)	Operating life of CO_2 sources, maximum storage capacity of sinks, and carbon footprint of compensatory energy	MILP with worst-case and robust stochastic formulations
(Zhang et al., 2018b)	Sink characteristics and investment limits represented by probability distributions	Two-Stage Stochastic MILP
(Leonzio et al., 2019a)	Selling price of CO_2-based chemical products	Stochastic MILP is specifically used with Monte Carlo Simulation
(Zhang et al., 2019a)	Carbon tax policy fluctuations (price)	Two-Stage Stochastic MILP uses Monte Carlo Simulation and the ε-constraint method for multi-objective optimization
(d'Amore et al, 2019)	Geological uncertainty in sequestration capacity and transport cost	Two-Stage Stochastic MILP using the Mersenne Twister algorithm
(Leonzio et al., 2020b)	Production cost variability of CO_2-based products (±20% from baseline)	Two-Stage Stochastic Programming with Monte Carlo Simulation
(Abdoli et al., 2023)	Oil yield and depletion rates revealed after investment decisions (decision-dependent uncertainty)	Multistage Stochastic Programming with endogenous uncertainty
(Nie et al., 2023)	Storage capacity, injection cost, and emission factors	Stochastic MILP is specifically used with Monte Carlo Simulation
(Shirazaki et al., 2024)	CO_2 emission rate and flue gas composition	Scenario-Based Stochastic Programming combined with Minimax Regret Robust Optimization
(Lei et al., 2024)	Storage/utilization facility disruptions	Two-Stage Stochastic MILP with scenario analysis and tanker transfer strategy
(Olson and Yaw, 2025)	Storage capacity and reservoir performance uncertainty	Stochastic MILP within *simCCS* framework using Monte Carlo sampling and heatmap aggregation
(Wu et al., 2025)	CO_2 injectivity (probabilistic) and economic parameters (interval)	Inexact Dynamic Source–Sink Matching (SSM) Model integrating Stochastic Mixed-Integer Programming (MIP), Chance-Constrained Programming (CCP), and Interval Linear Programming (ILP)

accuracy (Leonzio et al., 2019a). Similarly, Nie et al. adopted a stochastic MILP integrated with Monte Carlo Simulation to quantify uncertainties in storage capacity, injection costs, and emission factors, producing probabilistic cost ranges and confidence intervals for robust CCUS deployment planning (Nie et al., 2023). Shirazaki et al. combine scenario-based stochastic programming with robust optimization to minimize worst-case outcomes in CO_2 emission variability (Shirazaki et al., 2024), while Olson and Yaw integrate Monte Carlo sampling and heatmap aggregation within the simCCS framework to design infrastructure resilient to storage uncertainty (Olson and Yaw, 2025).

The development of CCS/CCUS optimization models has evolved from basic stochastic programming toward more advanced Two-Stage Stochastic MILP frameworks, allowing structured decision-making under uncertainty. Zhang et al. modeled probabilistic uncertainties in sink characteristics and investment limits, optimizing infrastructure under geological and financial variability (Zhang et al., 2018b). Building on this, Zhang et al. incorporated Monte Carlo Simulation and the ε-constraint method for multi-objective optimization, addressing trade-offs between cost, risk, and emission reduction under uncertain carbon tax policies (Zhang et al., 2019a). d'Amore et al. further refined the approach by applying the Mersenne Twister algorithm to simulate geological uncertainty in sequestration capacity and transport costs, enabling dynamic capacity adaptation across planning periods (d'Amore et al., 2019). Similarly, Leonzio et al. used Two-Stage Stochastic Programming with Monte Carlo Simulation to model market-driven variability in CO_2-based product costs, improving supply chain adaptability (Leonzio et al., 2020b). Most recently, Lei et al. extended the framework to resilience analysis, using scenario-based Two-Stage Stochastic MILP to manage storage and utilization facility disruptions through tanker transfer strategies (Lei et al., 2024).

Recent research has advanced beyond traditional two-stage stochastic approaches toward multistage and hybrid stochastic optimization frameworks to better capture dynamic and complex uncertainties in CCS and CCUS systems. Abdoli et al. extend a deterministic MILP for

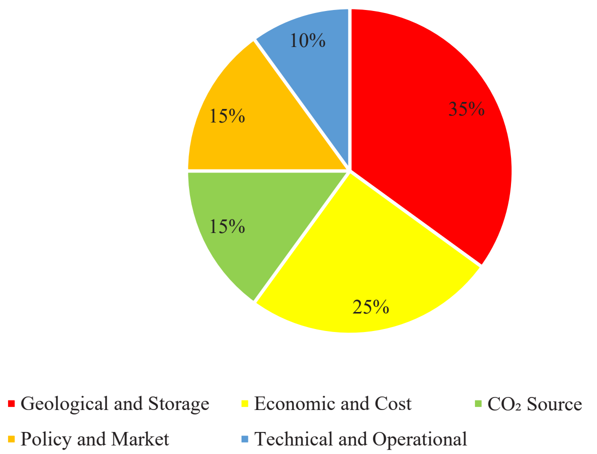

Fig. 14. Proportion of Uncertainty Categories in CCUS System Modeling.

CCS–EOR into a Multistage Stochastic Programming model with endogenous uncertainty, where key parameters—oil yield and depletion rates—are revealed only after investment decisions are made. This decision-dependent uncertainty creates a dynamic scenario tree, reflecting the feedback between planning actions and uncertainty realization, resulting in more realistic and cost-efficient strategies, with reported savings of approximately 8.8% compared to deterministic models (Abdoli et al., 2023). In a complementary direction, Wu et al. develop an Inexact Dynamic Source–Sink Matching (SSM) Optimization Model that integrates Stochastic Mixed-Integer Programming (MIP), Chance-Constrained Programming (CCP), and Interval Linear Programming (ILP) to jointly address probabilistic and interval uncertainties. This hybrid model captures stochastic variability in CO_2 injectivity through probability distributions while representing cost, demand, and economic parameters as intervals, providing a comprehensive framework for dynamic CCUS cluster planning (Wu et al., 2025).

Fig. 14 illustrates the proportional distribution of uncertainty types identified across the reviewed CCS and CCUS optimization studies. Five main categories of uncertainty were observed, each reflecting different dimensions of system complexity and decision-making risk:

- Geological and storage uncertainty (35%) is the most frequently addressed, encompassing factors such as uncertain storage capacity, injectivity, permeability, porosity, and reservoir performance (d'Amore et al., 2019; Zhang et al., 2018b; Abdoli et al., 2023; Olson and Yaw, 2025; Wu et al., 2025).
- Economic and cost uncertainty (25%) is the second most common, reflecting variations in production, transportation, and investment costs, as well as CO_2-based product prices (Zhang et al., 2018b; Leonzio et al., 2019a, 2020b; Nie et al., 2023; Wu et al., 2025).
- CO_2 source uncertainty (15%) arises from fluctuations in emission rates, flue gas composition, and the operational lifespan of emission sources (Shirazaki et al., 2024; Yi-Jun et al., 2014).
- Policy and market uncertainty (15%) stems from dynamic carbon pricing mechanisms and emission reduction policies that shape project feasibility and investment priorities (Zhang et al., 2019a; Nie et al., 2023).
- Technical and operational uncertainty (10%) involves potential disruptions to storage or utilization facilities, pipeline failures, and variability in enhanced oil recovery yields (Abdoli et al., 2023; Lei et al., 2024).

6. Conclusion

This review highlights the vital role of mathematical modeling in optimizing CCS/CCUS supply chain networks. It presents a structured overview of quantitative approaches, covering modeling scopes across CCS/CCUS components. Models are categorized as single-objective (SO)—typically focused on cost minimization—or multi-objective (MO), which may also incorporate environmental impacts. Deterministic models rely on fixed parameters, while non-deterministic models address uncertainty. The review also distinguishes between two-echelon (source–sink matching) and multi-echelon supply chain structures. Two-echelon models aim to minimize transport costs between emission sources and storage sites, whereas multi-echelon models incorporate capture technologies and transport infrastructure, providing a more comprehensive representation of CCUS systems.

The analysis review shows that the scope of mathematical modeling in the CCUS supply chain generally covers key components such as the CO_2 source or capture plant, the CO_2 sink (either storage or utilization), and the volume of CO_2 flow. More advanced developments focus on optimizing the transportation network. However, other important aspects—such as CO_2 pricing and policy, environmental impacts, social factors, resilience, and risk—are still rarely addressed.

The structural model analysis shows that most of the objective functions used focus on minimizing the total cost of CCS/CCUS systems, reflecting a strong emphasis on economic efficiency in current modeling practices. In models that apply a multi-objective function, additional objectives typically include minimizing environmental impact, cost-related risks, or social impacts.

The majority of models employ a single-objective function, while only a smaller portion use multi-objective approaches. In terms of model structure, many adopt a two-echelon framework, while the rest use more complex multi-echelon designs. Regarding uncertainty, most models are deterministic, with a smaller number incorporating uncertainty or being non-deterministic.

Uncertainty in CCS/CCUS modeling falls into five key categories. Geological and storage uncertainty is the most addressed, involving unknowns in storage capacity and reservoir behavior. Economic uncertainty covers cost fluctuations in production, transport, and CO_2-based product pricing. CO_2 source uncertainty relates to variable emission rates and source lifespans. Policy and market uncertainty stems from changing carbon prices and regulations. Technical and operational uncertainty includes risks like infrastructure failures and inconsistent enhanced oil recovery performance.

Therefore, taking into account the above conclusions, we propose several opportunities for future research in:

- Modeling scope: Expanding the scope by incorporating aspects such as CO_2 pricing and policy, environmental impacts, social factors, system resilience, and risk analysis.
- Model structure: Developing models with multi-objective functions and integrating uncertainty to better reflect real-world complexities. The uncertainty dimension could be addressed using less commonly applied approaches such as fuzzy programming and stochastic dynamic programming.

CRediT authorship contribution statement

Anna Maria Sri Asih: Writing – review & editing, Supervision. **Bertha Maya Sopha:** Writing – review & editing, Supervision, Conceptualization. **Mujammil Asdhiyoga Rahmanta:** Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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